



Structural investigation of ternary La/alkaline earth phosphate ($\text{La}_{(1-x)}\text{M}_x\text{P}_3\text{O}_y$) (M = Ba, Ca, Sr) glasses

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ABSTRACT

The structure of La-based metaphosphate glasses modified with aliovalent alkaline earth elements (M = Ba, Ca, Sr), with P:(M + La) ratio kept approximately at 3, were analyzed using Raman and IR spectroscopy. Increasing the concentration of the alkaline earths monotonically decreased the glass density and glass transition temperature, and shifted some vibrational modes by as much as 40 wavenumbers. The Raman frequencies of the *symmetric* POP and the P=O vibrations depended on the M/La ratio, but not on the type of alkaline earth addition; however, the type of alkaline earth cation significantly changed the *symmetric* PO₂ vibration. The average cation-oxygen coordination number remained near ~7 except for a slight decrease for high concentrations of Ca and Sr substitutions. The POP/PO₂ intensity ratio increased with increasing alkaline earth content. The replacement of divalent elements for trivalent La in a metaphosphate glass structure was partially accommodated by incorporation of protons as charge compensating defects. The protons were found to be in close proximity to the divalent cations.

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1. Introduction

La-based phosphate glasses have been widely studied for their interesting optical and thermal expansion properties [1–4], and more recently have been shown to be proton conductors as a bulk glass [5], as glass ceramics [6], and *via* grain boundary films in polycrystalline materials [7]. Alkaline earth (AE) phosphate glasses have long been known to conduct protons as well [8,9]. However, little is known about the structure and properties of ternary rare-earth-AE metaphosphate glasses, though related materials have also been shown to have potential as proton conducting electrolytes [10].

Because of the increased cation field strength and larger degree of phosphate chain cross-linking, the trivalent metaphosphate composition ($\text{M}^{3+}\text{P}_3\text{O}_9$), where the O:P ratio is 3, typically has higher T_g and is more resistant to aqueous attack compared to both the divalent metaphosphate ($\text{M}^{2+}\text{P}_2\text{O}_6$) and ultra-phosphate (O:P < 3)

compositions, and consequentially have greater potential for use in higher temperature applications [11–14].

Phosphate glass structures are often characterized by the bond types of the oxygen of the PO₄ tetrahedra within the glass. An oxygen can be covalently bonded to a phosphorus on an adjacent phosphate tetrahedron (bridging oxygen: BO) forming a POP bond; be coordinated to a metal cation (sometimes referred to as a non-bridging oxygen: NBO) forming a mixed ionic-covalent bond (POM); or form a P=O bond (π -bond) as a terminal oxygen (TO). Pure, vitreous P₂O₅ contains linked phosphate tetrahedra with 3 BOs, 1 TO, and no POMs. As cations are added to a P₂O₅ glass, POP bonds are destroyed in favor of POM bonds and the cation becomes a network modifier, and corner-sharing PO₄ chains or rings are formed. At the same time, the addition of modifying cations may delocalize the π -bond, making the distinction between the TO and the adjacent POM unclear [15]. In LaPO₄, for example, first principles computations indicated that the PO bonds of the isolated phosphate tetrahedra were of an equivalent type [16].

For glasses near the metaphosphate composition, most of the phosphate tetrahedra are in a configuration where two of its

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oxygen are involved in either bonding to the cation (POM bond), or forming a TO (a π -bonded P=O), while the other two form POP bonds that connect tetrahedra in a chain-like, polymeric structure. Each metal cation is coordinated with either the TO or the non-bridging oxygen of surrounding phosphate tetrahedra. The review by Brow and the references therein provide a detailed discussion of these structures [17].

In the binary AE metaphosphate glasses, the metal cation has an average oxygen-cation coordination number (CN) between 6 (Ca) [18] and 8 (Ba) [19], whereas for the trivalent lanthanum metaphosphate glasses the CN is closer to 6.5 [1].

In a mixed cation system the glass structure must reconfigure itself to accommodate the variation in radius and valence of the substituting cations. One such example of reconfiguration was observed in the Cu-ultra-phosphate glass system [20].

In the present paper, vibrational spectroscopy is used to investigate the effects of the addition of divalent cations to the stable lanthanum metaphosphate glass. As will be shown, the findings indicate that the replacement of La by lower-valence cations, while approximately maintaining the trivalent metaphosphate composition, does not create new types of TO bonds, and involves the incorporation of protons.

2. Experimental procedures

The trivalent metaphosphate stoichiometry, $P/(M + La) = 3$, was maintained approximately by replacing lanthanum with alkaline earth elements (Ca, Sr, Ba). Glasses of target composition $[La_{(1-x)}M_x]P_3O_9$ were synthesized with $x = 0, 0.2, 0.4, 0.6$, and 0.8 , where x is the replaced fraction of La, and y is the final oxygen content partly determined by the charge neutrality requirement.

The glasses listed in Table 1 were prepared by mixing high purity (99.999%) lanthanum oxide (Aldrich, CAS# 1312-81-8 – weighed in a glove box) and metal carbonates (ABCR 99.996% $SrCO_3$, CAS# 1633-05-2, Ventron 99.9975% $CaCO_3$, CAS # 471-34-1, Alfa Aesar 99.95% $BaCO_3$, CAS# 513-77-9), with 85% phosphoric acid (Merck VLSI Selectipur) directly into a Pt crucible. The crucible was covered and heated in air to a temperature $\sim 700^\circ C$ above the T_g of the glass. The glass was kept at temperature for up to 1.5 h, and then was poured on to a preheated brass plate ($\sim 250^\circ C$), quickly pressed with a room temperature brass plate to obtain a thin glass specimen, and further allowed to cool to room temperature. A Pt crucible was selected to prevent incorporation of Al or Si into the samples, as is the case with silica or alumina crucibles [3,21]. X-ray diffraction (Philips PW3020) did not detect any crystalline phases.

Due to the possible loss of P_2O_5 during glass melting, inhomogeneities in the melt, and unknown oxygen stoichiometry, the final compositions were analyzed by inductively coupled plasma mass spectrometry (ICP MS at the Max Planck Institute for Metal Research-Stuttgart). Finely crushed glass samples were analyzed for La, M^{2+} , and P, while bulk samples were analyzed for O to minimize contamination of the glass powders from atmospheric H_2O [22]. The error in mass percent is $\pm 0.3\%$ for La and Sr, and $\pm 0.5\%$ for P and O.

The clear, colorless, as-cast glasses were measured for density (AccuPyc 1330 He-pycnometer to $\pm 0.02 g/cm^3$), and for glass transition temperature using a differential scanning calorimeter (Setaram DSC 121) in flowing Argon. Samples were heated at $10^\circ C/min$ from RT to above the anticipated T_g , cooled to $300^\circ C$, and then heated above the T_g a second time; the reproducibility in T_g was $\pm 8^\circ C$ following this procedure.

The glasses were examined using single monochromator Raman Spectroscopy with a CCD detector (Jobin Yvon/Dilor, micro Raman version 'labram 010', 1 mW power at 632.817 nm (with Filter D 0.6), 20 μm diameter spotsize at 20x), as well as FTIR (Bruker IFS 66, with Globar Lamp, DTGS detector). Wavenumbers were accurate to $4 cm^{-1}$.

FTIR specimens were prepared in two ways. To observe the phosphorus structure at wavenumbers $< 1600 cm^{-1}$, samples were finely crushed then immediately mixed with KBr, and measured as quickly as possible to minimize the effects of exposure to atmospheric water. Spectra were measured in air in the $0-5000 cm^{-1}$ range. Because of the low intensity of the signal and overlap of KBr background in the high wavenumber range, thin bulk samples were prepared with a surface finish of $< 1 \mu m$ by diamond polishing, to observe the presence of OH stretches in the $> 1600 cm^{-1}$ range.

The Raman peaks were modeled using software developed at Max Planck Institute. The best fit for the POP main peak consisted of 3 sub-peaks, but the peak location was simply chosen as the highest value of the convoluted, overall peak. The *symmetric* PO_2 stretch was modeled as a Fano peak, and the *asymmetric* PO_2 stretch as a Lorentzian. All spectra were then background-subtracted, and normalized to the *symmetric* PO_2 peak for comparison.

It should be noted that water uptake during the melt may be affected by the temperature and time of melting [8]. Different La:M ratios required different melting temperatures, between 1150 and $1300^\circ C$, but identical melting conditions were used for fixed La:M ratios. It is expected that this may account for some of the deviation between samples. It may be assumed that the protons found in the glass are provided both from the water in the

Table 1

Target and actual compositions. T_g , molar volume, and density of La-Phosphate glasses. M is the alkaline earth element, with C, S, and B, series representing calcium, strontium, and barium, respectively. L1 represents the unsubstituted La-metaphosphate glass.

Sample	Target composition (moles)			Actual composition [moles (by ICP)]				T_g ($^\circ C$)	Molar volume (cm^3/mol)	Density (g/cm^3)
	La	M	P	La	M	P	O:P			
L1	25	0	75	24.51	0	75.49	2.96	620	124.4	3.31
C1	20	5	75	18.56	5.26	76.18	2.90	562	118.2	3.09
C2	15	10	75	13.68	10.12	76.21	2.89	534	116.6	2.95
C3	10	15	75	9.14	15.61	75.25	2.87	502	110.2	2.81
C4	5	20	75	4.59	19.98	75.43	2.92	500	110.4	2.62
S1	20	5	75	18.06	5.34	76.61	2.85	566	120.9	3.16
S2	15	10	75	13.54	10.78	75.68	2.78	544	115.5	3.11
S3	10	15	75	8.62	15.69	75.69	2.94	507	112.7	3.02
S4	5	20	75	4.25	20.12	75.63	2.93	456	113.5	2.94
B1	20	5	75	18.33	4.95	76.72	3.01	568	123.1	3.24
B2	15	10	75	13.39	9.33	77.28	2.95	509	123.6	3.27
B3	10	15	75	8.96	13.63	77.41	2.90	465	124.6	3.23
B4	5	20	75	5.60	18.43	75.97	2.98	454	119.9	3.22

starting materials (primarily in the H_3PO_4), and from the ambient atmosphere.

3. Results

3.1. General properties

The ICP MS analysis shown in Table 1, shows the starting composition and actual compositions, as well as the glass transition temperatures, molar volumes, and densities.

The molar volume was calculated using MW/ρ , where the MW of each glass was estimated from the ICP results assuming a LaP_3O_9 (MW = 375.82 g/mol) stoichiometry as a unit structure with adjustment for the percent of La replacement. As shown in Fig. 1, a monotonic decrease in T_g and density is observed with increasing AE substitution.

3.2. Raman spectroscopy

Fig. 2 shows Raman spectra for the La–Sr series, which are also representative of the other glasses, with assignments after Brow et al. [2], except the referenced $\nu_{\text{as}} \text{PO}_2$ is actually taken to be the P=O in our system, as is discussed in Section 4.2. Furthermore, be-

cause the $\nu_{\text{s}} \text{PO}_2$ signal is created by the stretch of a PO^- that is coordinated with the cations, and thus comprises a POM bond, POM is used in place of $\nu_{\text{s}} \text{PO}_2$ from here on. In this case only, the ‘M’ in POM does not distinguish between La and M^{2+} . With increased AE replacement, all peaks become broader, and the ratio of the symmetric POP peak to POM peak intensity increases as well. The results are summarized in Table 2.

With the addition of M^{2+} , the POP and POM ($\nu_{\text{s}} \text{PO}_2$) peak positions shift to lower values ($697\text{--}671.8 \text{ cm}^{-1}$ and $1189.5\text{--}1174.4 \text{ cm}^{-1}$, respectively), whereas P=O shifts to higher wavenumbers ($1258\text{--}1290.1 \text{ cm}^{-1}$). POP and P=O peak locations are plotted vs. M/La ratio in Fig. 3 and are found to be relatively independent of the type of the replacing AE cation.

In contrast, a plot of the $\nu_{\text{s}} \text{PO}_2$ (POM) peak frequency as a function of the M/La ratio reveals a dependence on the replacement cation type as shown in Fig. 4. Here, the wavenumbers decrease with the addition of AE elements, with the decrease most pronounced for Ba (from 1189.5 to 1170.4 cm^{-1}), and least for Ca (to 1178.7 cm^{-1}).

The intensity ratio of the POP peak to POM peak is shown in Table 2. There is a general increase in the ratio as expected, and a slight dependence on AE cation type is noted, particularly at higher substitutional levels. Literature values for the divalent Ba- and Ca-metaphosphate (Takabe et al. [24], and Bertoluzza et al. [25]) are included in Table 2 for comparison; note, however, that differing values for these same materials were reported by Nelson and

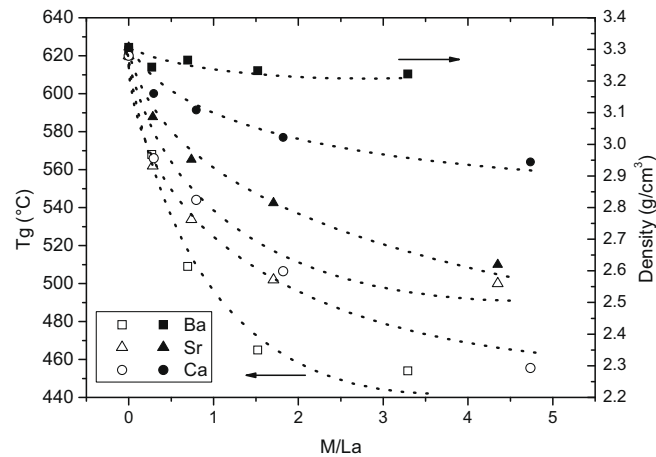


Fig. 1. Density and T_g of the glasses as a function of modifying metal cation. Open symbols are the glass transition, and the closed represent the density.

Table 2
Raman peak locations of ν_{s} POP, POM, P=O , as well as actual and predicted POP/POM intensity ratios.

Sample	Raman peak locations (cm^{-1})			Predicted POP/POM	Actual POP/POM
	POP	POM	P=O		
L1	697.0	1189.5	1241.5	0.52	0.49
C1	687.7	1185.5	1256.9	0.60	0.48
C2	676.4	1183.7	1269.0	0.66	0.53
C3	673.4	1181.8	1276.4	0.67	0.65
C4	667.4	1178.7	1290.8	0.72	0.83
S1	687.6	1185.2	1258.0	0.63	0.50
S2	679.3	1183.4	1266.4	0.64	0.54
S3	673.7	1179.1	1280.9	0.69	0.66
S4	671.8	1174.4	1290.1	0.73	0.78
B1	687.7	1183.3	1258.4	0.63	0.47
B2	674.7	1178.8	1271.4	0.70	0.56
B3	674.0	1174.8	1276.7	0.76	0.59
B4	671.5	1170.4	1285.4	0.73	0.64
BaP ₂ O ₆ [23]	690	1180	1250	–	0.48
CaP ₂ O ₆ [24]	693	1178	1277	–	0.51

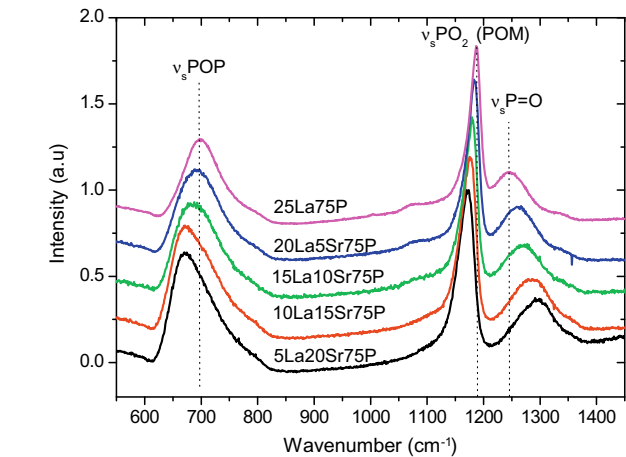


Fig. 2. Raman spectroscopy of La–Sr metaphosphate glasses. Starting metal mole amounts shown. The spectra are representative of the La–Ba and La–Ca glasses as well.

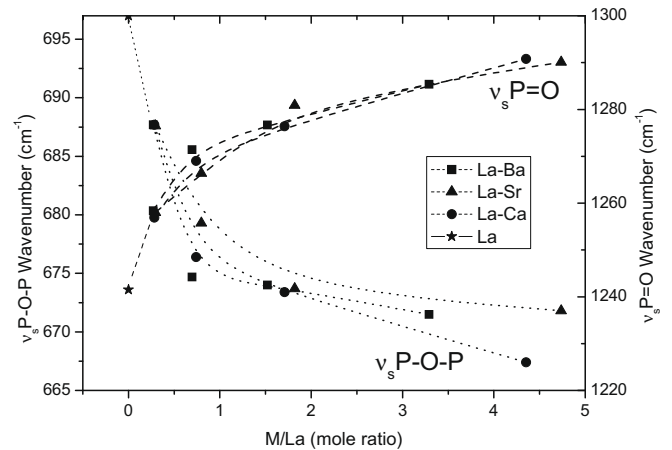


Fig. 3. POP and P=O peak locations as a function of alkaline earth addition.

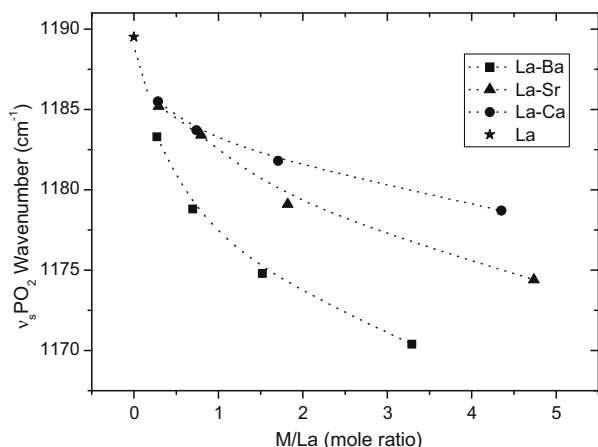


Fig. 4. ν_s PO₂ peak locations as a function of the type and amount of alkaline earth addition.

Exharos [23]). As the background varied slightly from sample to sample, care was taken in background subtraction and normalizing the data by the same procedure across all spectra. The POM peak was chosen for normalization as it is the peak with the highest intensity, the most defined peak, and its frequency shifts the least of all peaks across the compositional range.

3.3. FTIR

In the $<1600\text{ cm}^{-1}$ range, similar trends are observed with FTIR POP and PO₂ peaks, and are thus not shown. In the $>1600\text{ cm}^{-1}$ range shown in Fig. 5, as also noted by Efimov and Pogareva [26], several OH stretches are observed, though the overlap of the phosphate backbone in this range makes the absorption at 2900 cm^{-1} the most useful. The absorption increases with increasing AE addition.

After Abe and Shimakawa [8], using Lambert–Beer's Law, $A = \epsilon ct$, where A is the absorption, ϵ is the molar absorbance coefficient, c is the concentration of absorbing species, and t is the thickness of the sample, the concentration of protons in the glass may be estimated from:

$$[\text{H}^+] = 2 \frac{\log(T_{3600}/T_{\text{OH}})}{\epsilon_{\text{H}_2\text{O}} t} \quad (1)$$

The results are shown in Table 3, where T_{3600} and T_{OH} are the transmittance at 3600 cm^{-1} , and the transmittance at the OH absorption wavenumber, respectively. A correlation between the

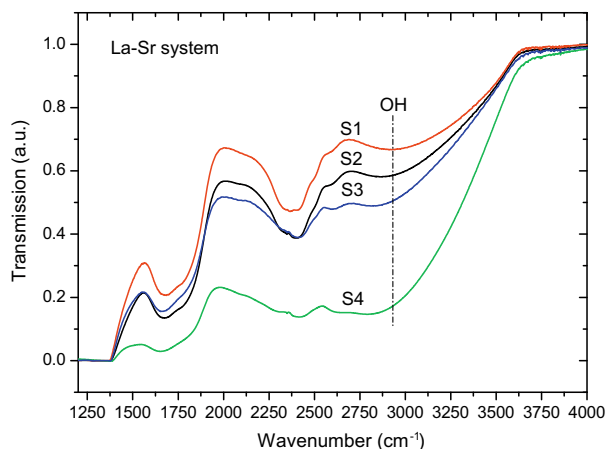


Fig. 5. FTIR on thin solid samples of the La–Sr metaphosphate glasses.

addition of AEs and proton concentration is observed. Anomalies in the La–Ca and the M/La ~ 4 series are noted, and are probably due to inhomogeneities in the melt. The mole concentration of protons in the glass is roughly estimated using the molar volume of the glass, and assuming the protons in the glass have same absorptive properties as bulk water.

4. Discussion

4.1. General properties

The glass transition temperature, T_g , and density, ρ , decrease monotonically with increasing La replacement, suggesting no abrupt transitions in structure. These changes can be primarily associated with a decrease in cation field strength and cation mass, respectively. The lower T_g of Ca than Sr for the highest substitution amount may in part be explained by the higher water content found in sample C4.

4.2. Asymmetric PO₂ vs. P=O Raman stretch

We have assigned the stretch at $\sim 1250\text{ cm}^{-1}$ to the P=O stretch instead of to an asymmetric PO₂ stretch, as shown in Ref. [2]. The oxygen involved in the PO₂ stretch are directly bonded to adjacent metal cations and thus should be dependent on the type of cation in the glass, as is evident from the change in peak frequency of the symmetric PO₂ stretch in Figs. 3 and 4. A P=O stretch for the same base structure of glass, on the other hand, is expected to be independent of cation type because it does not form strong bonds with neighboring metal cations. An analysis of our data shown in Figs. 3 and 4 shows that the peak frequency of the 1250 cm^{-1} stretch is independent of cation type. If this were the asymmetric counterpart to the symmetric PO₂ stretch shown in Fig. 4, both frequencies should be expected to show a dependence on cation type.

It is interesting to note that for certain compositions, there is overlap of the asymmetric PO₂ with the P=O vibration, as shown in Fig. 12 of Brow's review paper [17], though historically the literature shows inconsistent assignments of this peak. In our case the evidence suggests the 1250 cm^{-1} signal to be dominated by the P=O vibration.

Also, in the Na-metaphosphate glass, the intensity of this peak appears to be particularly affected by water content as pointed out by Brow [17]. By comparing two different works, Brow shows that the intensity of the 1250 cm^{-1} peak is severely decreased when glasses are formed in the presence of water. This observation is consistent with the assignment of that peak to a P=O stretch rather than an asymmetric PO₂ stretch, because a proton is more likely to affect the intensity of the terminal oxygen stretch rather than a metal–oxygen stretch.

We also note that though the TO is thought to coordinate with a metal cation as described by Hoppe [15], it appears from the independence of this Raman stretch on cation type, that in our case this bonding to the cation is weak compared to the interactions of the metal with non-terminal oxygen atoms.

4.3. Raman intensities

The ICP results show that the overall O:P ratio of 3 is approximately maintained even for high replacement of La^{3+} by the divalent AEs cations. Both divalent ($\text{M}^{2+}\text{P}_2\text{O}_6$) and trivalent ($\text{M}^{3+}\text{P}_3\text{O}_9$) metaphosphate glasses by themselves have a POP/PO₂ Raman intensity ratio near 0.5, whereas in this case substituting divalent AE cations for La^{3+} with near-constant $\text{P}/(\text{M} + \text{La}) = 3$, the POP/PO₂ Raman intensity ratio increases, especially in the La–Ca series (Table 2).

Table 3

Summary of approximate proton content of glasses.

Sample	M/La	Molar absorbance coefficient (l/mol cm)	[H ⁺] (mol/l)	Mol% H ⁺	Mol% M ²⁺	H/M
L1	0	132	0.05	0.54	0	–
C1	0.283	129.6	0.30	5.4	5.26	1.0
C2	0.740	127.2	0.38	6.9	10.1	0.7
C3	1.708	124.8	0.16	2.5	15.6	0.2
C4	4.352	122.4	0.73	14.5	20.0	0.7
S1	0.295	130.1	0.20	2.5	5.22	0.5
S2	0.796	128.2	0.23	2.7	10.7	0.2
S3	1.820	126.3	0.39	4.5	15.6	0.3
S4	4.736	124.4	0.88	10.0	20.0	0.5
B1	0.270	131.6	0.16	1.9	4.95	0.4
B2	0.696	131.2	0.38	4.7	9.33	0.5
B3	1.521	130.8	0.60	7.5	13.6	0.5
B4	3.291	130.4	0.44	5.2	18.4	0.3

In principle, the increase may be predicted if two considerations are made. The first, is that as compared to a mixture of two metaphosphate of similar chain structure (divalent: P/M = 2, and trivalent P/M = 3), our system has excess phosphate because we keep the P/(M + La) ratio of ~3. Over the pure, divalent metaphosphate, the 'excess' P in the ternary system is simply equal to the amount of divalent metal cation, i.e., $P_{xs} = M^{2+}$.

The second consideration is the assumption that POM bond concentration remains constant upon addition of the divalent AE ions, because otherwise this would require a change of the oxygen coordination sphere around the metal cations, as is discussed in Section 4.4.

With AE replacement for La in LaP_3O_9 , charge neutrality would require a reduction in the O:P ratio from 3 (for no La replacement) to 2.85 (for 80% La replacement). This would result in a composition that is usually designated as ultra-phosphate. A slight decrease for some compositions is indeed observed, however; the error on the measurement does not allow for a quantitative analysis of this value. The O:P values less than 3, cannot be accounted for by experimental error, but instead represent a loss of oxygen from the system. In an anhydrous, oxygen-deficient environment, an ultra-phosphate composition would be formed through creating extra POP bonds. In environment where excess oxygen and hydrogen is present, as is the case of our glasses using H_3PO_4 as a starting material, the short melt times prevent significant loss of oxygen and phosphorous, and with the presence of excess of protons, is thought to maintain a structure closer to a metaphosphate than an ultra-phosphate, if charge neutrality can be assumed. A decrease in oxygen stoichiometry (i.e., formation of an ultra-phosphate), from a defect chemistry standpoint, is formally described by forming an oxygen vacancy (or POP bond) for every two M^{2+} 'substitutions', or for every two excess phosphorus. Thus, with AE substitution, the POP/POM ratio is expected to increase, and a simple model may be used to predict the expected POP/POM intensity ratio. The predicted intensity ratio is then given by the equation:

$$\frac{I_{POP}}{I_{POM}} = \frac{1}{2} + \left(\frac{P - 3L - 2M}{4(L + M)} \right), \quad (2)$$

where L , M , and P are the corresponding values listed in Table 1. This equation assumes that the intensity of the POM stretch does not change with AE substitution. The expected and measured POP/POM intensity ratios for Sr, Ca, and Ba substitutions are shown in Table 2 are plotted in Fig. 6.

In this comparison, the actual POP content follows a similar trend, but falls consistently below the expected one, except for the highest additions of Ca and Sr. The discrepancy may be due to several factors, such as decreased metal–oxygen coordination (thus a decrease in POM intensity), or incorporation of water that dissociates POP bonds and produces protonic defects. The effects of bond polarizability have not been taken into account.

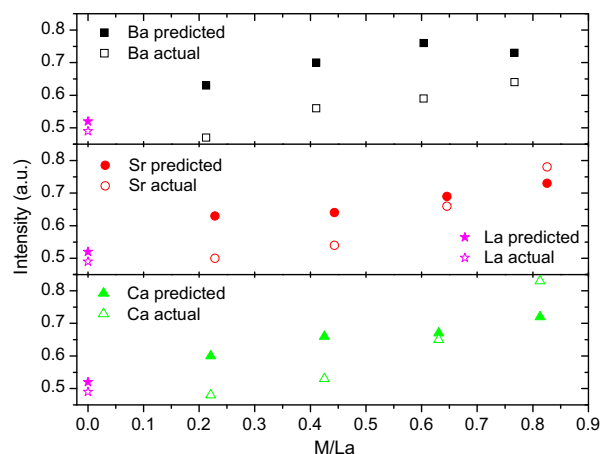


Fig. 6. Observed POP/POM Raman intensity ratio, with expected values based on the glass compositions.

4.4. Metal–oxygen coordination

According to the studies by Hoppe et al. [1,19], the oxygen coordination for a La should be 6–7 in the metaphosphate glass, and should be 8 for a Ba-metaphosphate. Matsubara showed that Ca was in 6-fold coordination [18]. It is assumed that Sr would adopt a CN between Ba and Ca given their cation potentials. It is difficult to predict the coordination number from this study; however, examination of the ν_3PO_2 peak frequency vs. the mean cation field strength (z/r), reveals a near linear correlation, as shown in Fig. 7.

The linear correlation indicates that the coordination numbers for Sr and Ba do not change significantly from the La coordination, otherwise the ionic radius, as determined by Shannon [27], would be changing, and deviations from linearity would show. For Ca and Sr, the trend does deviate with increasing substitution, suggesting the coordination may be slightly decreasing from 7, as would be expected for a Ca-metaphosphate. This finding may in part explain the higher than predicted POP/POM ratios for C4 and S4 in Fig. 6. There are various other models for predicting the vibrational frequencies of cations, as discussed by Nelson [23], though none fit the symmetric PO_2 peak.

4.5. Protons

The trend in deficiency of POP bonds for all other compositions may also be related to the observation by FTIR of protons in the system. Besides formation of a POP bond, positive charge deficit created by incorporation of an M^{2+} into the trivalent metaphosphate may be balanced by incorporation of protons. Indeed, the

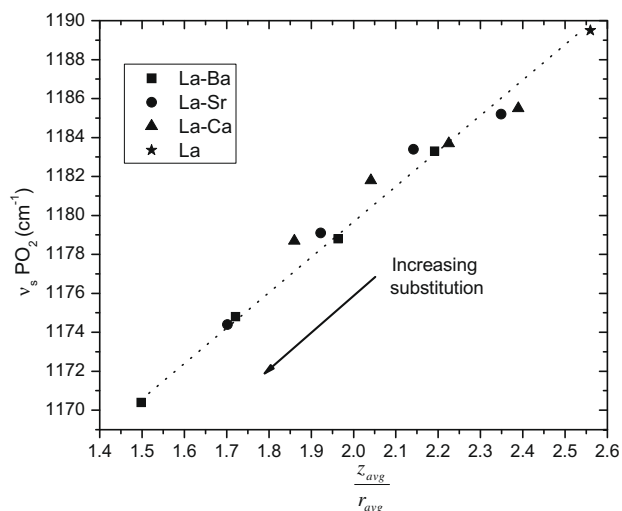


Fig. 7. $\frac{z_{avg}}{r_{avg}}$ vs. symmetric PO_2 (POM) Raman stretch. Radius in 7-fold coordination.

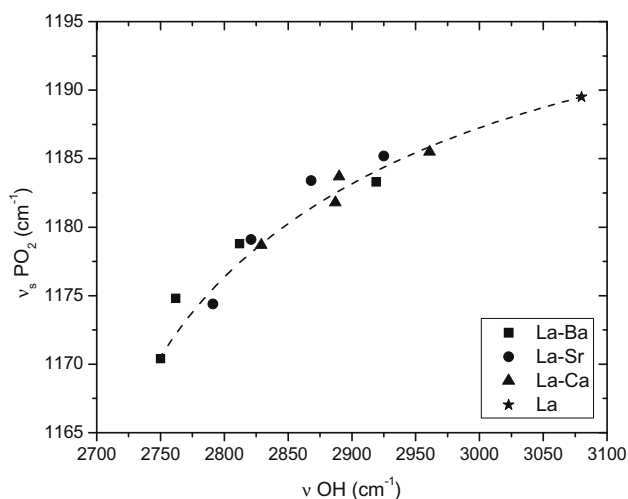


Fig. 8. Correlation between Raman $\nu_s \text{PO}_2$ (POM) stretch and FTIR νOH stretch.

FTIR data (Table 3) shows increasing $[\text{H}^+]$ with increasing M^{2+} addition, although the ratio H/M shows considerable variation. The origin of this variation is in part due to the approximation of Eq. (1), and the model assumptions leading to Eq. (2), and also exemplifies the difficulty of controlling the proton content during synthesis of the glasses.

Assuming that a ‘trivalent’ metaphosphate glass structure dominates when the $\text{P}/(\text{La} + \text{M})$ is ~ 3 , then charge-compensating protons may be expected to reside in the vicinity of modifying divalent cations. Then, a correlation should be expected between the frequencies the POM stretch and the vibration of one of the OH peaks. This can be verified by comparing the FTIR and Raman data, as is shown in Fig. 8.

A strong correlation is indeed found between the location of the broad OH stretch around $\sim 2900 \text{ cm}^{-1}$ and the symmetric PO_2 (POM) stretch. This supports the postulate that protons reside as a charge compensating defects in the immediate vicinity of the modifying divalent cations, making these act as weak proton traps. The exact bonding configuration requires further investigation, though it is expected the configuration is as $\text{P}-\text{O}-\text{M}-\text{O}-\text{H}-\text{O}$ or $\text{P}-\text{O}-\text{M}-\text{O}-\text{P}-\text{O}-\text{H}-\text{O}$, so that the POM stretch is related to an OH

stretch in close proximity to the M. While the present observations pertain to room temperature, where the vibrational spectra were collected, the degree of such proton trapping and its effect on proton conductivity at elevated temperatures in mixed RE–AE metha-phosphates remains to be evaluated.

5. Conclusions

The density and glass transition temperature are smoothly decreased from the pure LaP_3O_9 metaphosphate glass with the addition of AE elements while keeping $\text{P}/(\text{La} + \text{M}) \sim 3$. The ternary glasses maintain predominantly a trivalent metaphosphate-type structure despite the addition of lower valence AE elements. The structure configures itself by increasing POP/POM ratio and the incorporation of charge-compensating protons. Proton concentration correlates with the amount of alkaline earth substitution in the Lanthanum metaphosphate glass. The indications are that these protons reside in the immediate vicinity of the divalent metal cation, which can act as proton traps.

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